Python GMSH INP Converter Short Guide

Read Original Article by Dip28 here:
http://dip28p.web.fc2.com/calculix/gmshinpconvert/index.html

(or in PDF copy in the folder: \c4w\Hlp\conv\GmshInp_by_Dip28.pdf)

NOTE
THIS CONVERTER IS FOR ORIGINAL GMSH INP FILE (not of GMSH by Prool, which save in native CalculiX format and don't need to be converted)

Short Description

1. Create the geometrical model in GMSH (v>2.5.1)
2. Save as Abaqus .INP ("Save All" option, Fig. 1)
3. Use Converter to transfer in CalculiX compatible .INP file (Fig. 2,3)

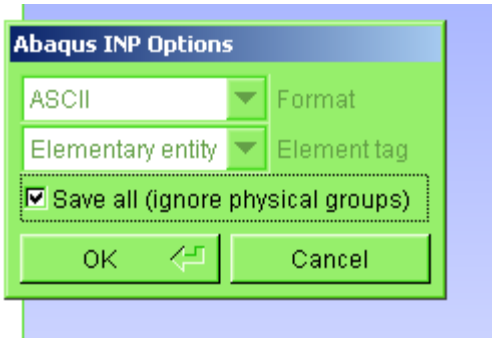


Fig. 1 - Saving INP file

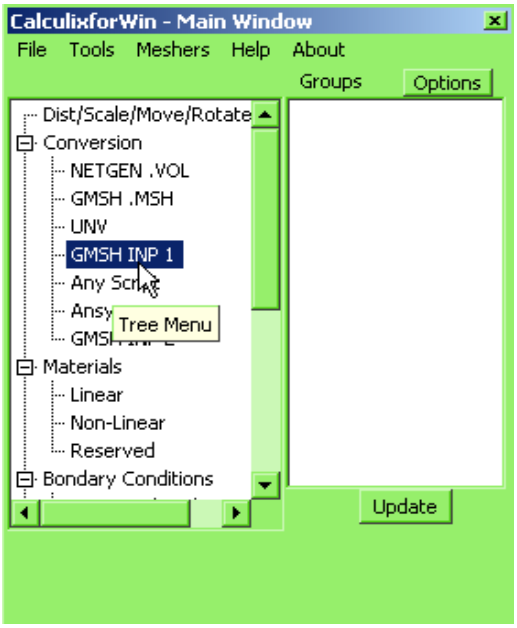


Fig. 2 Running GMSH INP converter

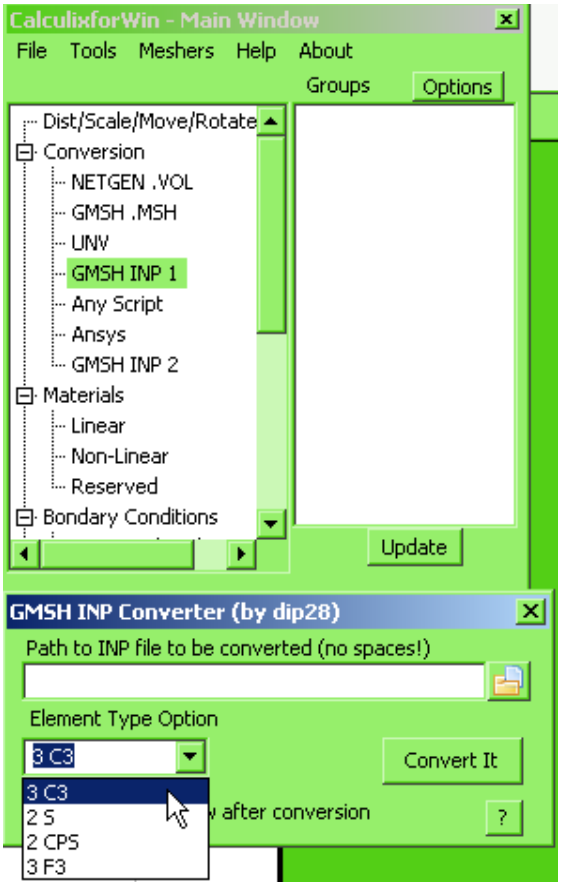


Fig. 3 - Set path to .INP file (no spaces, no Cyrillic letters!)
Choose Element Type option (3 C3 for bodies, 2 S for shells, etc)
Push "Convert It" and wait when python script is done

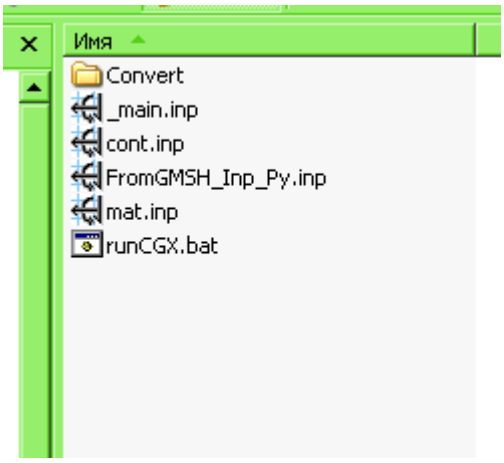


Fig. 4 - The result of conversion "FromGMSH_Inp_Py.inp" file in the working folder

The example (provided by dip28)

Copy the code of geometry below

```
cl1 = 5;
Point(1) = {10, 0, 0, cl1};
Point(2) = {50, 0, 0, cl1};
Point(3) = {100, 0, 0, cl1};
Line(1) = {1, 2};
Line(2) = {2, 3};
Extrude {0, 10, 0} {
Line{1, 2};
}
Extrude {0, 0, 20} {
Surface{6, 10};
}
Physical Surface(183) = {31};
Physical Surface(184) = {49};
Physical Volume(181) = {1};
Physical Volume(182) = {2};
```

Run GMSH, go Geometry>Edit File.

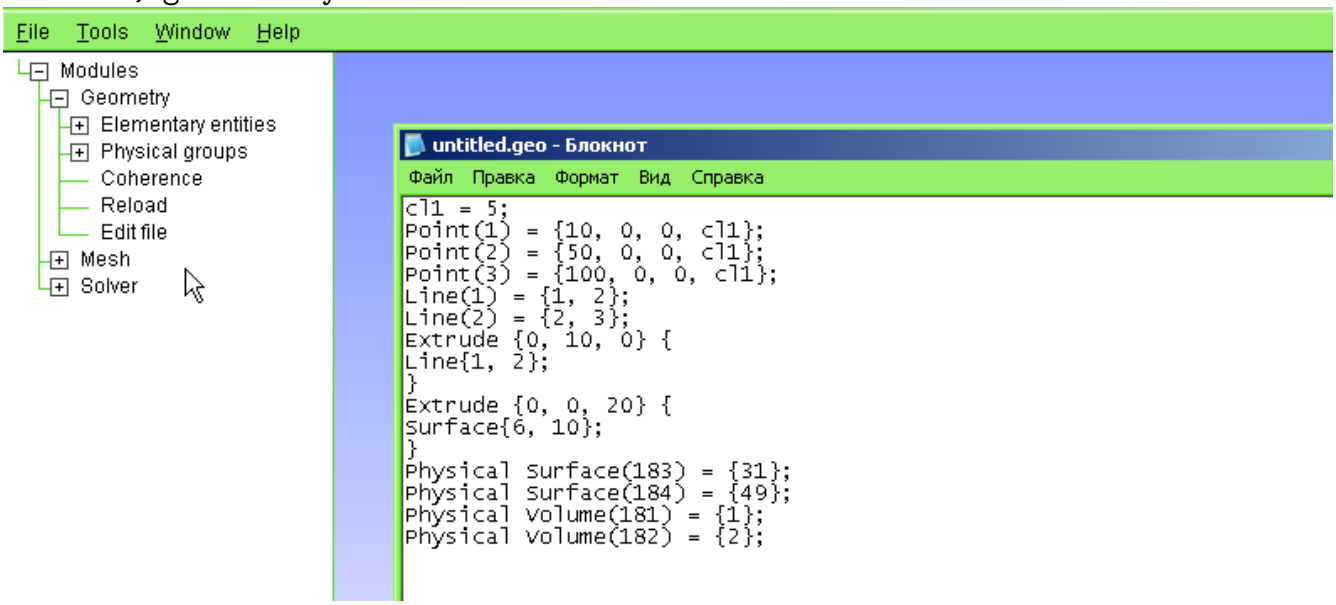


Fig. 5 - Paste the code into the untitled.geo and save it.

(There is important note about using text editor in GMSH. When the code isn't correct the program can crashes while opening.

Go to C:\Documents and Settings\...\Application Data\ and delete (or correct) untitled.geo file)

Press reload

Go to Mesh, press "3d", then "Set order 2"

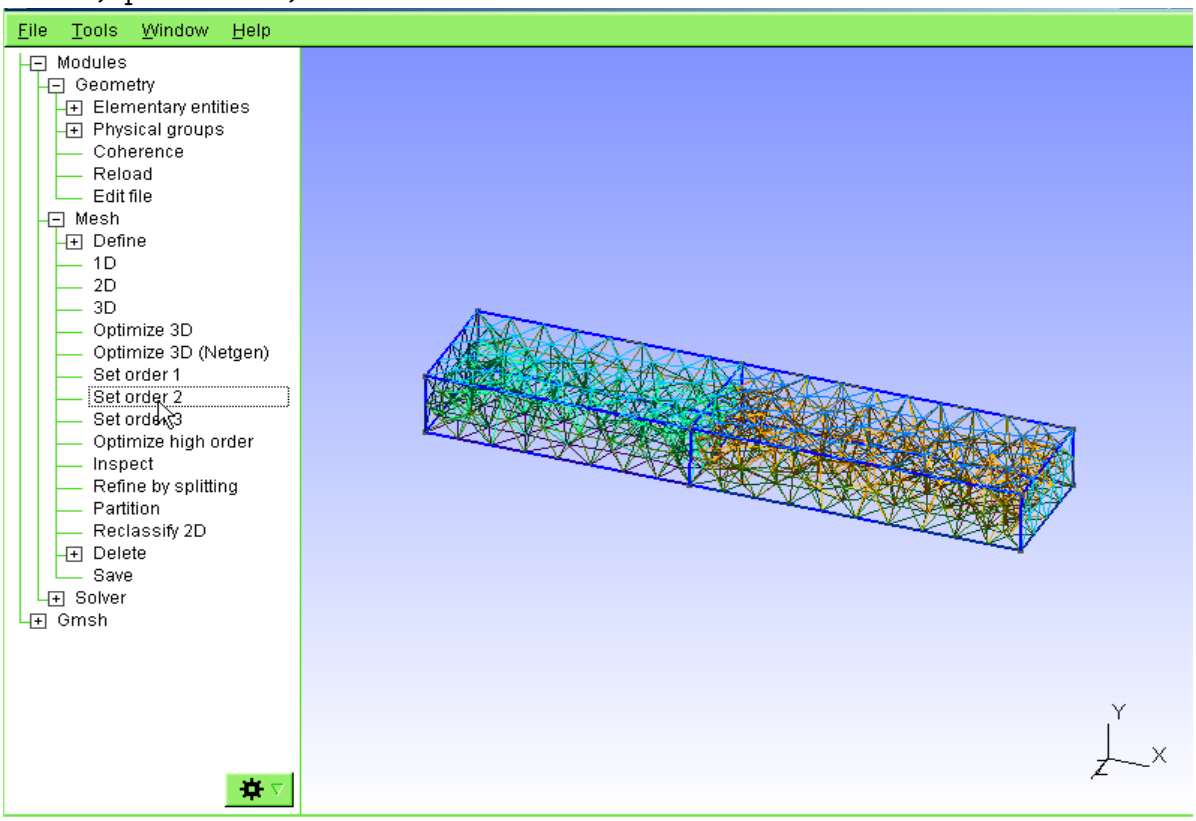


Fig. 6

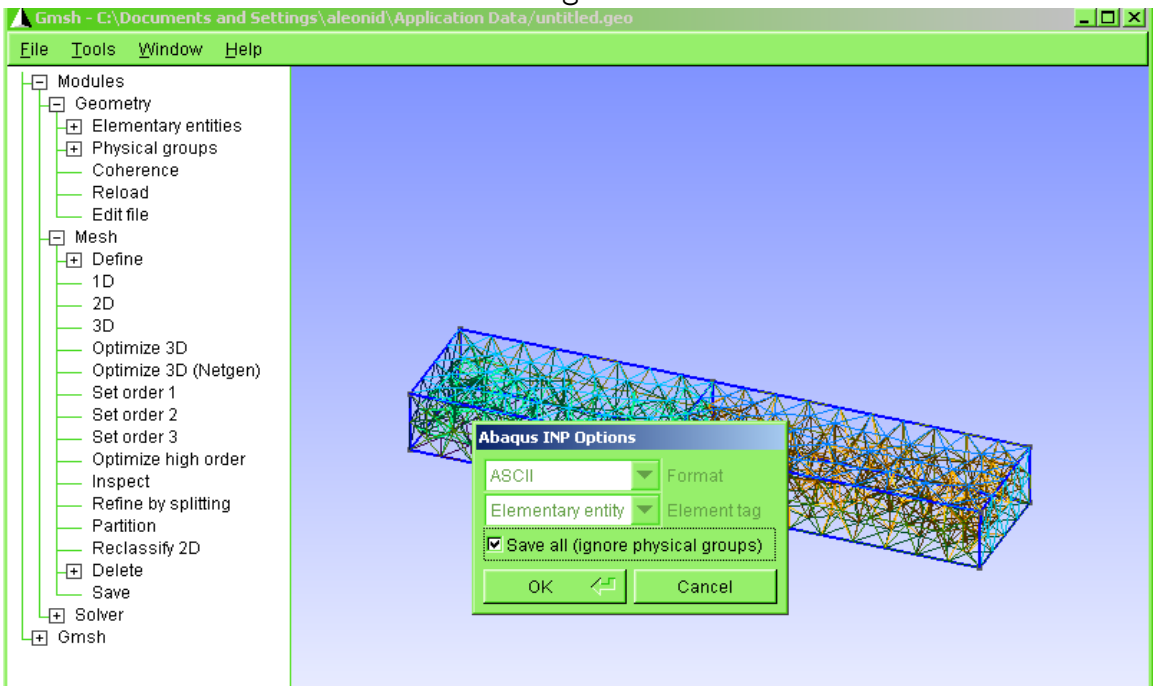


Fig. 7 - Save file as INP file

Run CalculiXForWin

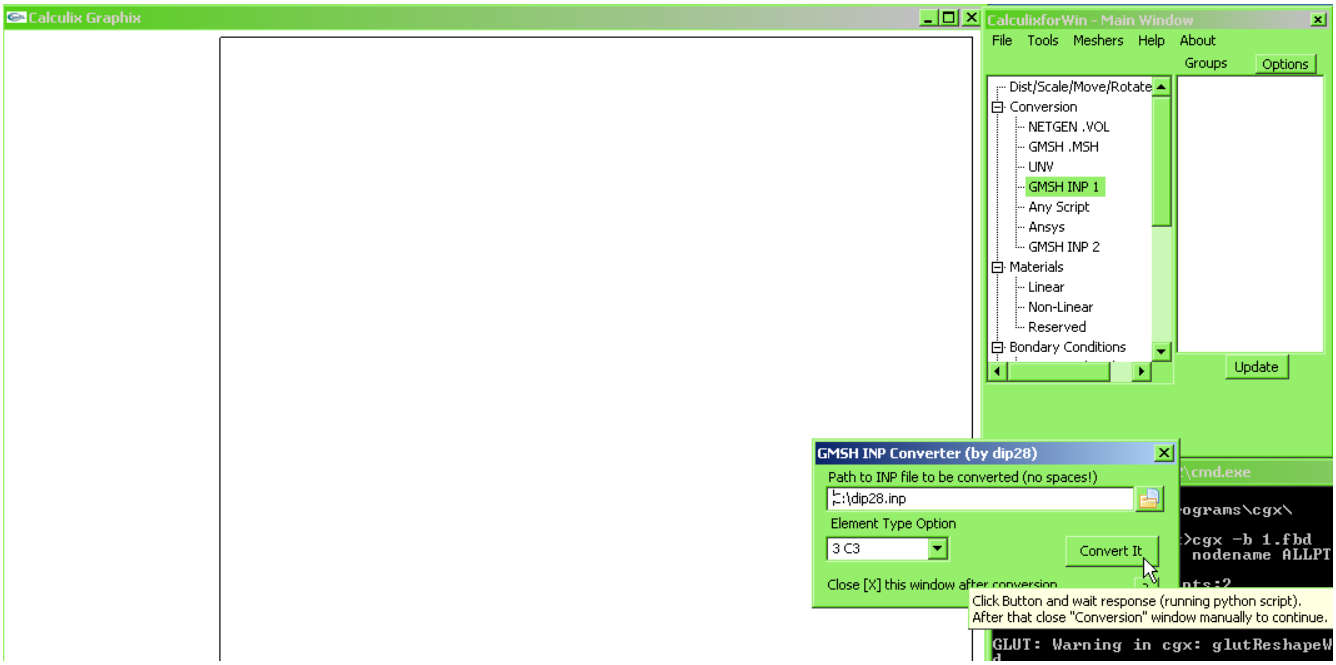


Fig 8 - Set the path to the inp file and press "Convert It"

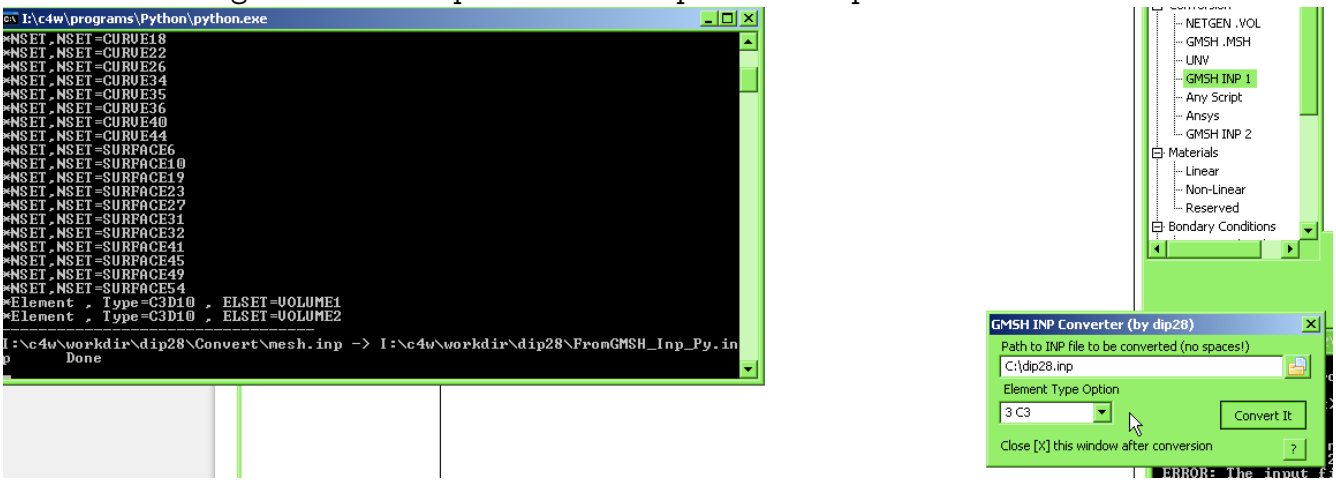


Fig. 9 When conversion is done, close "GMSH INP" window manually and go the the CalculiXForWin Menu >File>Open CCX INP

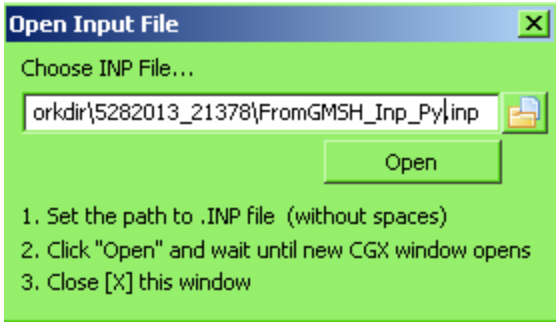


Fig. 10 Choose the converted file in the working folder and press open. New CGX window will be opened.

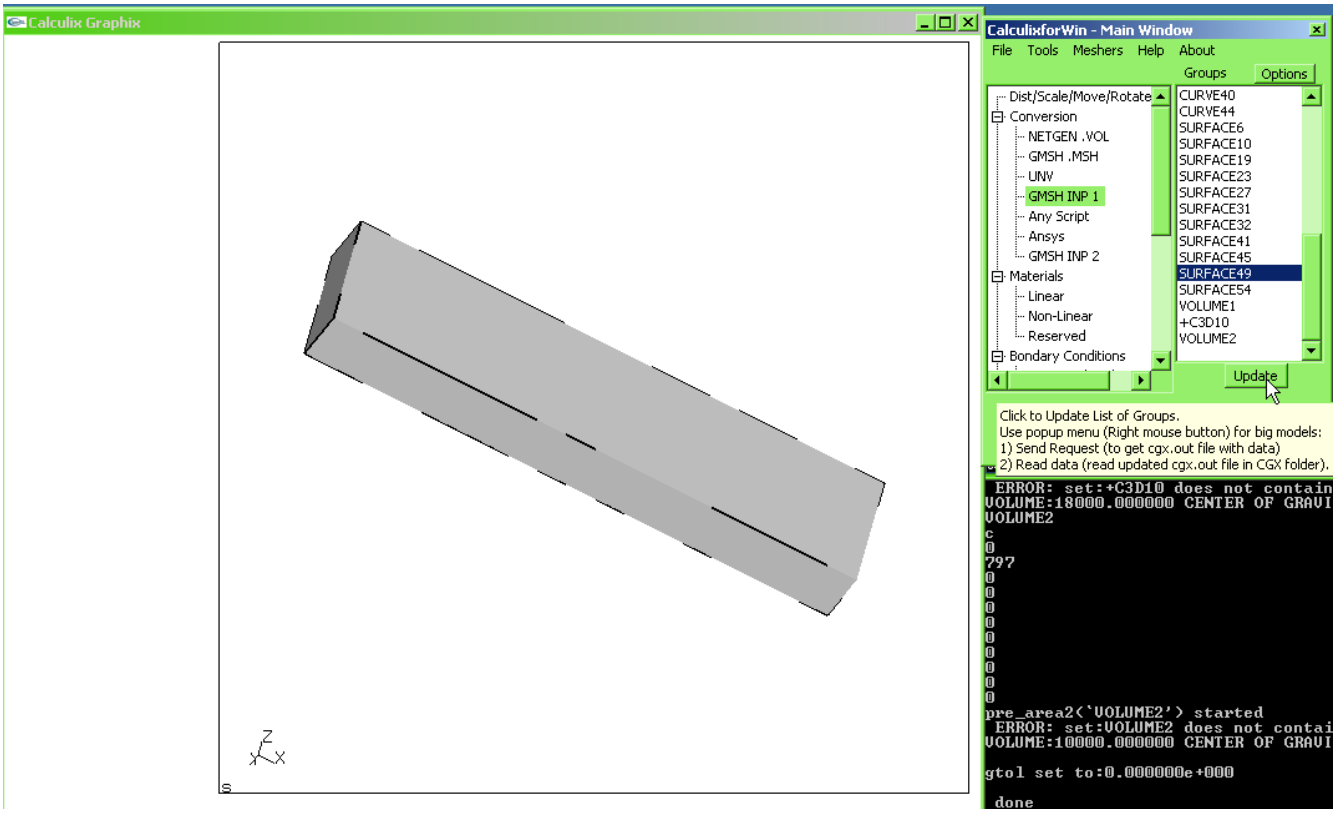


Fig. 11 - Press "Update" button under the list box and see the list of groups.

We don't need to scale the model in this case because we use native GMSH geometry (not STEP file). But you can use "qdis" card to check the dimensions for any case.